TECHNOCREDITS 9765463742

Java

Day 1
1) Installation & importance of path variable
2) Why should we write class in java (To fulfil business requirements)
3) Variables & datatypes (String & int)
4) Methods/functions
   a. Method name
   b. Parameters (Arguments)
   c. Return type
   d. Method body
5) Object creation
6) Entry point function : main() method
7) Code compilation
8) Compiler's responsibilities
   a. Check syntactical errors
   b. Add extra code or optimization
   c. Convert .java file to .class file(Byte code)
9) Interpreter - for execution of code
10) Will do first practical program

Day 2
1) Parameters/ Arguement
2) Local & Instance
3) Instance variables have their default values.
4) Filename should match classname [B.P]
5) One java file can have multiple class [.class file will be created for every class]
6) One java file should have only one class with exact matching name

Day 3
1) 8 Primitive datatypes with default values:
   a. byte [1 byte]
   b. short [2 bytes]
   c. int [4 bytes]
   d. long [8 bytes]
   e. float [4 bytes]
   f. double [8 bytes]
   g. character : char [2 bytes]
   h. true & false : boolean [1 byte]
2) Memory required for each datatype
3) Range of each datatype
4) No default values will be assigned to the local variable.
5) A local variable has to initialize before we use it.

Day 4
1) Conditional statement if - else
2) If else ladder
3) || and && condition
4) Basic usage of == and equals() method
   a. For String comparison use equals() method
   b. For any data type(int, float, double, char, short, byte and boolean) comparison use == operator

Days 5
1) Nested If else

Day 6
1) For Loop :
   a. Loops are used to execute a set of statements repeatedly until a particular condition is satisfied.
   b. If the number of iteration is fixed, it is recommended to use for loop.

Notes:-
   a. In for loop, initialization happens first and only one time, which means that the initialization part of for loop only executes once.
   b. Condition in for loop is evaluated on each iteration, if the condition is true then the statements inside for loop body gets executed. Once the condition returns false, the statements in for loop does not execute and the control gets transferred to the next statement in the program after for loop.
   c. After every execution of for loop's body, the increment/decrement part of for loop executes that updates the loop counter.
   d. After every execution of for loop's body, the increment/decrement part of for loop executes that updates the loop counter.
   e. After third step, the control jumps to second step and condition is re-evaluated.

Day 7
1) Loop variation, while loop, break statement.

Loops are used to execute a set of statements repeatedly until a particular condition is satisfied.

While Loop:
   a. The Java while loop is used to iterate a part of the program several times. If the number of iteration is not fixed, it is recommended to use while loop.
   b. In while loop, condition is evaluated first and if it returns true then the statements inside while loop execute. When condition returns false, the control comes out of loop and jumps to the next statement after while loop.

Note: The important point to note when using while loop is that we need to use increment (or decrement) statement inside while loop so that the loop variable gets changed on each iteration, and at some point condition returns false. This way we can end the execution of while loop otherwise the loop would execute infinitely.

Day 8
1) Eclipse introduction
2) While loop examples
3) Static non access modifier
4) Install Git and Create an account on GitHub.

Day 9 & 10
1) SSH-Key setup and Import project from GitHub

Day 11
1) Return type.

Day 12
1) Static - Non-Static

Day 13
1) Communication between classes.
2) Git PR: Two different ways

Day 14
1) Switch case

Day 15
1) JDK, JRE, JVM
2) Reference variable
3) How System.out.println() works
4) Scanner class

Day 16
1) Array
   a. Array Declaration
   b. Initialize or assign value to array element
   c. Access values from array using index
   d. Length is a property of an array - that will give total number of elements in array.

Day 17
1) String class methods
   a. int length()
   b. char charAt(int index)
   c. String trim()
   d. int lastIndexOf(char ch)
   e. String toUpperCase()
   f. String toLowerCase()
   g. String replace(char oldchar, char newchar)
   h. String replace(charsequece oldchar, charsequece newchar)
   i. String replaceFirst(String oldString, String newString)
   j. String replaceAll(String oldString, String newString)

2) Scanner class

Day 18
1) Debugging

Day 19
1) String class method -> indexOf

Day 20
1) Array comparision
2) Wrapper class
3) PrimeNumber
4) Package
5) Import

Day 21
1) Seperate dgits from number
2) Access modifer

Day 22
1) Encapsulation

Day 23
1) Inheritance(Part 1)

Day 24
1) Inheritance(Part 2)

Day 25
1) Inheritance(static & final methods)

Day 26
1) Final Keyword
2) Inheritance Interview questions
3) Inheritance (protected)
4) Multilevel Inheritance
5) Multiple Inheritance
6) Polymorphism
   a. Overloading
   b. Overriding

Day 27
1) Overview of OOPS
   a. Encapsulation
   b. Inheritance
   c. Polymorphism
      a. Overloading
      b. Overriding
   d. Abstraction:
      Abstraction can be achieved in two ways.
      a. Interface
      b. Abstract class

Day 28
1) Abstract Class
2) Difference between Interface & Abstract

Day 29
1) Abstraction Interview
2) CoVerientReturnType
3) TernaryOperator

Day 30
1) Abstraction(Java8 Features)
2) ClassCastException
3) Assignment 26 Solution

Day 31
1) Constructor
2) Super
3) This.
4) Program: Char Count

Day 32
1) Abstract Class Has Constructor & Nonstatic block

Day 33
1) Static & Non-Static block
2) Constructor
3) Enum

Day 34
1) Blank final variable
2) Ascii value

Day 35
1) Singleton Design Pattern

Day 36
1) String pool
   a. There are two ways to create a String object.
   b. String class is immutable
   c. Detail discuss == & .equals()
2) toString() method
3) intern() method
4) Program: Replace second last occurrence

Day 37
1) StringBuffer
2) StringBuilder
3) What is Exception & Exception Handling

Day 38
1) Boolean class has constructor which can take String value, internally it is going to convert it in boolean by checking if value is "true" then assign true else assign false to boolean variable.

for example, when we say
Boolean b1 = new Boolean("Hi");
value of b1 will be false as string value assigned to b1 is not "true", anything apart from string "true" is assigned to boolean as false.

if we say
Boolean b1 = new Boolean("true");
b1 will get boolean value as true.

now in a give question,
Boolean b1 = new Boolean("Hi"); // internally false will be assigned to b1
Boolean b2 = new Boolean("Hello"); // internally false will be assigned to b2

b1 == b2; will be false as we are comparing referece and two objects always store at different location.
equal method will return true as b1 and b2 both having value false.

2) Points to be remember for Singleton Design pattern.

   a. Create private constructor - It will prevent to instantiate the Singleton class from outside the class.
   b. getConnection() should be static, this provides the global point of access to the Singleton object and returns the instance to the caller.
   c. Private static reference variable of Connection class.
   ie. private static connection con;
   it will get memory only once bez of static.

Day 39
1) Exception(try & catch)

Day 40
1) Exception(try, catch & finally)

2) Check different-different scenarios

   a. only try -> CE
   b. try & catch -> Allowed
   c. try multiple catch -> allowed [child then parent]
   d. try & finally -> allowed
   e. try & finally & catch -> CE
   f. try catch finally finally -> CE [only one finally block is allowed]
   g. nested try catch -> allowed
   h. try catch inside try -> allowed
   i. try catch inside catch -> allowed
   j. try catch inside finally -> allowed

3) System.exit(int status)
4) Predict output of below program.

```
 int m1(int x){
try{
 int y = 10/x;
 return y;
}catch(RuntimeException re){
 return 30;
}catch(Exception e){
 return 20;
}catch(Throwable t){
 return 40;
}finally{
 return 400;
}
//return 500;
}
```

m1(10); --> 400
m1(0); --> 400

Day 41
1) Exception(throw & thows)

Day 42
1) Exception(throws)

Day 43
1) Quick Revision(Throw-throws & Exception)
2) Custom-Exception
3) toCharArray & Arrays Class
4) ClassNotFoundException

Day 44
1) Array Limitation
2) Collection Overview
   a. List
      i) boolean add(Object obj)
      ii) Object get(int index)
      iii) int size()
      vi) boolean remove(Object obj) // Techno
      v) Object remove(int index) // 2
      vi) void add(int index, Object obj) add element at specific index
      vii) set(int index, Object obj) // replace an element at given index [3, Yash]
      viii) int indexOf(Object obj); // it will return index of an element
      ix) boolean contains(Object obj);
      x) void clear()

   b. Set

3) Coding Practice

Day 45
1) Collection(Map & HashMap)
2) What we have covered so far:
   a. Why collections framework [limitations of array]
   b. Collections framework hierarchy
   c. List interface property [allow duplicate, maintain insertion order]
   d. ArrayList initialization.
   e. ArrayList class & useful method (13 methods)
   f. ArrayList pass by reference
   g. Enhance for loop
   h. Convert Array to List
   i. asList method return non modifiable list
   j. Collect student references to a list
   k. fail fast and fail safe operations
   l. iterate a loop using Iterator
   m. Predicate example
   n. Compare two array list

Day 46
1) Collection (Pass collection reference to method) [ArrayList<Employee>]

Day 47
1) Collection(FailFast & FailSafe)

Day 48
1) Collection(HashSet Iterator)

Day 49
1) Collection (TreeSet)
2) Map (HashMap, LinkedHashMap, Hashtable, TreeMap)
3) Git Part
   a. How to see local branches
   b. how to see remote branches
   c. How to see all branches
   d. Reset
   e. Reset & checkout diff
   f. How to delete untracked files
   g. How to delete local & Remote Branch

Day 50
1) ArrayList & LinkedList

Day 51
Holiday Utilies

Pattern printing - 1.
https://drive.google.com/open?id=1J4y_Z25UV7S0NJKqOklOFGCntT355nqK

Pattern printing - 2.
https://drive.google.com/open?id=1uirxt2g4rZEBV_68uiln-31y-mR3OaZ_

Day 52
1) Upload project on gitHub

Day 53
1) Git stash

Day 54
1) Github
   a. How to give access of your repo.
   b. How to create branch on github and start working on it.
   c. How to rename Local and Remote Branch
      Local Branch rename : git branch -m <New_branch_name>
      Remote branch rename:
      git push origin --delete <old_branch_name>
      git push origin -u <new_name>
   d. How to delete Repo from github.

Day 55
1) How to create conflict & solve them

Day 56
1) HashMap Internals
2) Equals & Hashcode method Contract
3) Comparator & Comparable